# A Demand-Driven Framework for India's AI Infrastructure Planning: Linking Compute, Energy, Physical Resources, and Semiconductor Dependencies

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***Abstract*— National artificial intelligence (AI) strategies increasingly treat high-performance compute accelerators as primary instruments of technological sovereignty. However, compute scaling is physically bounded by electricity transmission, cooling water, semiconductor lithography, advanced packaging, and geopolitically concentrated supply chains. This paper develops an evidence-backed Five-Pillar Conceptual Framework integrated with an auditable, six-layer Demand-Cascade Model to evaluate India's AI infrastructure trajectory across 200,000, 500,000, and 1,000,000 graphics processing unit (GPU) deployment scenarios. Parameterized from empirical data centre audits, foundry specifications, and Central Electricity Authority telemetry, the model translates accelerator targets into peak electrical power, annual energy, cooling water withdrawal, front-end silicon wafer starts, 2.5D interposer packaging allocations, and High-Bandwidth Memory (HBM) stacks. For a baseline 500,000-GPU deployment, the model projects a peak facility power demand of 891.8 MW—equivalent to 56.6% of India's current operational colocation data centre capacity—and an annual consumption of 6.25 TWh. Crucially, the empirical results refute the conventional hypothesis that front-end semiconductor fabrication capacity is the binding physical chokepoint: required wafer starts represent merely 1.74% of global sub-5nm output, whereas 2.5D advanced packaging (CoWoS) consumes 7.41% of total worldwide capacity. Consequently, policy planning must invert standard priorities, subordinating premature domestic fab construction to power transmission corridors, geographical spatial diversification, and bilateral advanced packaging partnerships.**

***Keywords*— Artificial intelligence infrastructure, semiconductor sovereignty, demand-cascade modeling, power grid constraints, advanced packaging (CoWoS).**

## I. INTRODUCTION

Artificial intelligence (AI) has transitioned from an algorithmic discipline to an ultra-compute-intensive industrial apparatus governed by capital expenditure, high-voltage power grids, and physical microelectronic supply lines. Across leading economies, access to high-density compute clusters is viewed as indispensable for national security, economic productivity, and strategic autonomy [1]. In India, this ambition is codified in the statutory authorization of the IndiaAI Mission with a financial outlay of ■10,371.92 crore (~$1.25 billion USD) [1], complemented by the India Semiconductor Mission (Semicon India 2.0) with an expanded budgetary horizon of ■1,27,500 crore [2]. Policy pronouncements articulate deployment goals scaling from an initial 10,000 graphics processing units (GPUs) toward national deployments exceeding 100,000 to 500,000 accelerators by 2030.

However, prevailing policy discourse frequently decouples accelerator procurement from the physical and industrial systems required to sustain compute operations. Compute does not exist in the ether: an enterprise accelerator cluster demands continuous megawatt-scale electrical power, high-voltage substation interconnects, gigalitres of cooling water under tropical ambient temperatures, and complex supply chains spanning lithography, advanced packaging, and high-bandwidth memory (HBM). Without rigorous physical co-planning, fiscal subsidies risk procuring silicon that sits idle behind multi-year transmission queues or causes severe municipal resource conflicts.

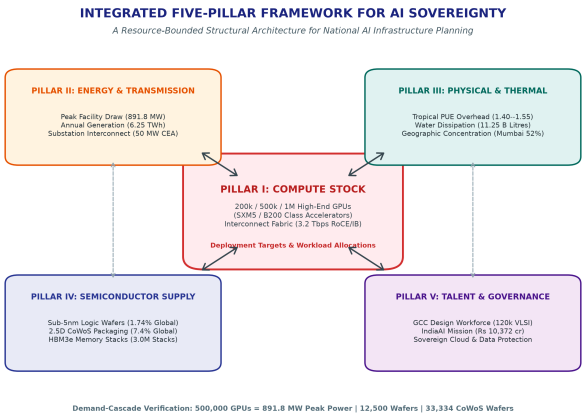



*Fig. 1. Integrated Five-Pillar Conceptual Framework for Sovereign AI Infrastructure Planning.*

To address these vulnerabilities, this paper introduces an integrated Five-Pillar Conceptual Framework (Fig. 1) and formulates an auditable, reproducible six-layer Demand-Cascade Model. By mapping exogenous accelerator targets (200k, 500k, 1M GPUs) through empirical physical equations, we quantify the boundary constraints governing India's AI infrastructure trajectory and identify where the true structural chokepoints reside.

## II. LITERATURE REVIEW AND RESEARCH GAP

Existing scholarship on AI infrastructure remains fundamentally bifurcated into two disconnected domains:

**A. Data Centre Thermodynamics and Grid Scaling:** Shehabi et al. [5] establish empirical power profiles for hyperscale accelerator architectures, documenting that server-level host overheads (CPUs, network interface cards, fans) multiply raw chip thermal design power (TDP) by a factor of 1.6 to 2.1. The International Energy Agency (IEA) [4] estimates that global data centre electricity demand could exceed 1,000 TWh by 2026, driven primarily by generative AI workloads. Kaufmann et al. [15] evaluate transmission congestion, highlighting that multi-hundred-megawatt compute clusters create local substation bottlenecks that cannot be resolved through generic generation capacity alone. In India, industry censuses by Colliers [6] and JLL [7] reveal an operational colocation capacity of 1,575 MW in mid-2026, but document an acute spatial vulnerability: 52.4% of national capacity is concentrated in Mumbai, where grid interconnection queues now extend to 18–36 months under Central Electricity Authority (CEA) guidelines [3].

**B. Semiconductor Geopolitics and Foundry Chokepoints:** Akimitsu et al. [14] document that over 90% of commercial sub-5nm logic fabrication resides in Taiwan under TSMC [12]. Furthermore, telemetry from SemiAnalysis [9] and TrendForce [8] highlights that the primary hardware bottleneck for modern AI accelerators is not raw wafer lithography, but Chip-on-Wafer-on-Substrate (CoWoS) 2.5D advanced packaging. Similarly, Counterpoint Research [10] demonstrates that global HBM3e supply is concentrated in a tight duopoly between SK Hynix (~57%) and Samsung (~38%), creating severe geopolitical exposure. In upstream minerals, the USGS [11] notes that China controls 98% of global refined gallium production, a critical input for power electronics.

**C. The Research Gap:** Despite these detailed domain studies, there is a complete absence of an integrated, quantitative framework linking national GPU procurement targets directly to macro-grid capacity, water stress, and upstream semiconductor foundry/packaging allocations. Prior strategic policy papers either treat compute as an unconstrained commodity or present heuristic vulnerability scores that obscure underlying engineering mechanisms. This paper bridges that gap.

## III. DEMAND-CASCADE MODEL FORMULATION

The Demand-Cascade Model maps an exogenous target of *N* accelerators through six deterministic layers. All variables and empirical baselines are drawn from verified primary registers (Table I).

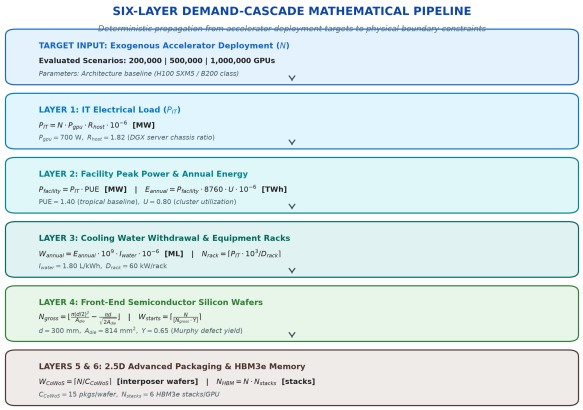



*Fig. 2. Mathematical Pipeline of the Six-Layer Demand-Cascade Model.*

**Layer 1: IT Electrical Load ($P_{IT}$):** Total active IT load accounts for accelerator board TDP ($P_{gpu}$) and server chassis overhead ($R_{host}$ = 1.82), including dual host CPUs, memory, 3.2 Tbps InfiniBand/RoCE network cards, and PCIe switches [13]:

$$P_{IT} = N \cdot P_{gpu} \cdot R_{host} \cdot 10^{-6} \quad [MW] \qquad (1)$$

**Layer 2: Facility Peak Power and Annual Energy:** Facility grid interconnection capacity incorporates Power Usage Effectiveness (PUE = 1.40 tropical baseline):

$$P_{facility} = P_{IT} \cdot PUE \quad [MW] \qquad (2)$$

Annual electrical energy consumption ($E_{annual}$ in TWh) is a function of cluster capacity utilization factor $U = 0.80$ across 8,760 annual operating hours:

$$E_{annual} = P_{facility} \cdot 8760 \cdot U \cdot 10^{-6} \quad [TWh] \qquad (3)$$

**Layer 3: Cooling Water & Racks:** Annual evaporative cooling water consumption reflects cooling intensity ($I_{water}$ = 1.80 L/kWh) [4]:

$$W_{annual} = E_{annual} \cdot 10^{9} \cdot I_{water} \cdot 10^{-6} \quad [ML] \qquad (4)$$

Physical rack footprint is determined by power density ($D_{rack}$ = 60 kW/rack):

$$N_{rack} = \lceil P_{IT} \cdot 10^{3} / D_{rack} \rceil \qquad (5)$$

**Layer 4: Semiconductor Wafer Starts:** Gross dies per 300mm wafer ($N_{gross}$) are calculated from radius r=150mm and die area $A_{die}$=814 $mm^2$ via the Murphy geometric equation:

$$N_{gross} = \lfloor \pi r^2 / A_{die} - 2\pi r / \sqrt{(2A_{die})} \rfloor = 63 \qquad (6)$$

Applying net functional defect yield Y=0.65, net good dies are $N_{good} = \lfloor 63 \cdot 0.65 \rfloor = 40$. Total required 300mm wafer starts are:

$$W_{starts} = \lceil N / N_{good} \rceil \qquad (7)$$

**Layers 5 & 6: 2.5D Packaging & HBM Memory:** Total 300mm silicon interposer wafers required for CoWoS packaging ($W_{CoWoS}$) are derived from finished packages per interposer wafer $C_{CoWoS}$ = 15:

$$W_{CoWoS} = \lceil N / C_{CoWoS} \rceil \qquad (8)$$

Total HBM3e memory stacks scale linearly with stack density ($N_{stacks}$ = 6 per GPU):

$$N_{HBM} = N \cdot N_{stacks} \qquad (9)$$

**TABLE I: EMPIRICAL PARAMETER REGISTER**

| ID | Symbol | Unit | Low | Base | High | Source |
|---|---|---|---|---|---|---|
| PAR-001 | P_gpu | Watts | 700 | 700 | 1000 | NVIDIA [13] |
| PAR-002 | R_host | ratio | 1.60 | 1.82 | 2.10 | LBNL [5] |
| PAR-003 | PUE | ratio | 1.25 | 1.40 | 1.55 | SBI / JLL [7] |
| PAR-004 | U | ratio | 0.65 | 0.80 | 0.90 | IEA [4] |
| PAR-005 | I_water | L/kWh | 0.20 | 1.80 | 2.50 | IEA [4] |
| PAR-006 | D_rack | kW/rack | 40 | 60 | 100 | IEEE TCPMT |
| PAR-007 | A_die | mm² | 650 | 814 | 1628 | TSMC [12] |
| PAR-008 | Yield | ratio | 0.55 | 0.65 | 0.75 | TrendForce [8] |
| PAR-009 | CoWoS | pkgs/wfr | 12 | 15 | 18 | TrendForce [8] |
| PAR-010 | HBM | stacks | 6 | 6 | 8 | Counterpoint [10] |

## IV. EMPIRICAL RESULTS AND DISCUSSION

**A. Scenario Analysis (200k, 500k, 1M GPUs):** Table II summarizes model outputs across all three deployment targets. In the baseline 500,000-GPU scenario, operational IT load reaches 637.0 MW. Under tropical climatic conditions (PUE = 1.40), peak facility interconnect requirement escalates to **891.8 MW**. This represents **56.62% of India's entire operational colocation data centre capacity** (1,575 MW in mid-2026). At 80% utilization, annual electricity consumption totals **6.25 TWh/year** (~0.38% of national gross generation), while evaporative water dissipation reaches **11,249.5 ML/year** (11.25 million cubic meters).

**TABLE II: SCENARIO CASCADE MODEL RESULTS**

| Scenario | Variant | IT (MW) | Fac (MW) | TWh/yr | Water (ML) | Wafers | CoWoS |
|---|---|---|---|---|---|---|---|
| 200k GPUs | Low | 224.0 | 280.0 | 1.59 | 318.9 | 4,445 | 16,667 |
| | Base | 254.8 | 356.7 | 2.50 | 4,499.8 | 5,000 | 13,334 |
| | High | 420.0 | 651.0 | 5.13 | 12,831 | 10,527 | 11,112 |
| 500k GPUs | Low | 560.0 | 700.0 | 3.99 | 797.2 | 11,112 | 41,667 |
| | Base | 637.0 | 891.8 | 6.25 | 11,250 | 12,500 | 33,334 |
| | High | 1,050.0 | 1,627.5 | 12.83 | 32,078 | 26,316 | 27,778 |
| 1M GPUs | Low | 1,120.0 | 1,400.0 | 7.97 | 1,594 | 22,223 | 83,334 |
| | Base | 1,274.0 | 1,783.6 | 12.50 | 22,499 | 25,000 | 66,667 |
| | High | 2,100.0 | 3,255.0 | 25.66 | 64,156 | 52,632 | 55,556 |

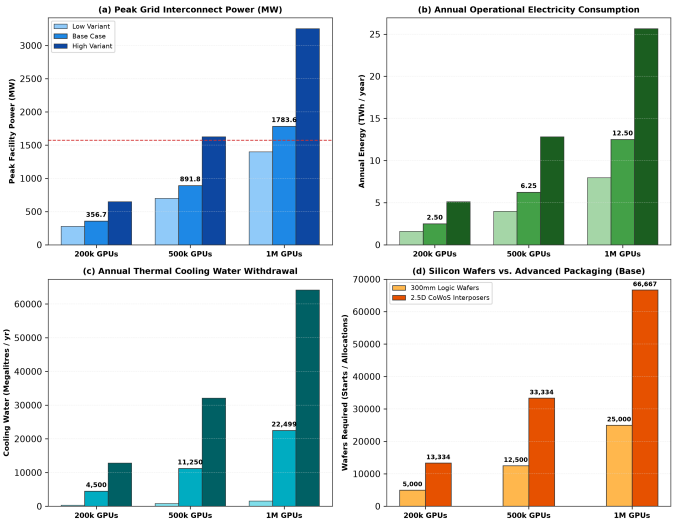



*Fig. 3. Physical Resource Demands across GPU Deployment Scenarios.*

**B. Refuting the 'Silicon Wafer' Hypothesis:** A foundational finding of this research is the empirical refutation of the widespread policy belief that domestic front-end semiconductor fabrication capacity is the binding physical chokepoint for sovereign AI compute:

- **Sub-5nm Logic Wafer Demand:** Deploying 500,000 GPUs requires exactly **12,500 300mm wafer starts**.
- **Global Foundry Supply Context:** TSMC's sub-5nm manufacturing capacity exceeds 60,000 wafer starts per month, or ~**720,000 wafers annually** [12].
- **Relative Scarcity:** India's 500k-GPU requirement consumes **merely 1.74% of annual global sub-5nm wafer output**. Even a 1,000,000-GPU deployment requires only 3.47%.

Thus, raw silicon lithography is easily absorbable by commercial foundry supply lines provided procurement capital is allocated.

**C. Identification of the True Structural Chokepoints:** The model demonstrates that the binding constraints reside in packaging, memory, and domestic infrastructure (Table III):

1) *Advanced 2.5D Packaging (CoWoS):* Packaging 500,000 GPUs requires **33,334 CoWoS interposer wafers**. With global CoWoS capacity bounded at ~420,000–450,000 wafers/year [8], India would absorb **7.41% of total worldwide packaging output**. This represents an acute global chokepoint characterized by 9–14 month supplier backlogs.

2) *High-Bandwidth Memory (HBM3e):* Deploying 500,000 accelerators requires **3,000,000 HBM stacks** (~6.67% of annual global output), where India possesses zero domestic fabrication or assembly.

3) *Spatial Transmission Congestion:* Supplying 891.8 MW into Mumbai (which holds 52.4% of existing data centres) requires 18 dedicated 50 MW CEA substations [3], triggering grid saturation and multi-year interconnection delays.

4) *Municipal Hydrological Strain:* Consuming 11.25 billion litres/year under conventional wet cooling creates severe conflict with urban drinking water.

**TABLE III: BOTTLENECK DISAGGREGATION (500K GPUS)**

| Component | 500k Need | Global Capacity | Share | Strategic Severity |
|---|---|---|---|---|
| Front-End Wafers | 12,500 wfrs | 720,000 /yr | 1.74% | Low Scarcity |
| 2.5D CoWoS Pkg | 33,334 wfrs | 450,000 /yr | 7.41% | Critical Chokepoint |
| HBM3e Stacks | 3.0M stacks | 45.0M /yr | 6.67% | Monopoly Risk |
| Colocation Power | 891.8 MW | 1,575 MW (Ind) | 56.6% | Severe Grid Strain |
| Cooling Water | 11,250 ML | Local Basin | High | Local Conflict Risk |

**D. Sensitivity Analysis:** Tornado analysis (Fig. 4) reveals that facility power is exceptionally sensitive to PUE: transitioning from tropical air cooling (PUE = 1.60, 1,019.2 MW) to direct liquid cooling (PUE = 1.20, 764.4 MW) **saves 254.8 MW of peak grid demand** at 500k scale. Wafer requirements vary by +17.6% under early defect yields (55%), while large-die reticle scaling (1628 mm² dual-die Blackwell) more than doubles wafer starts to 26,316.

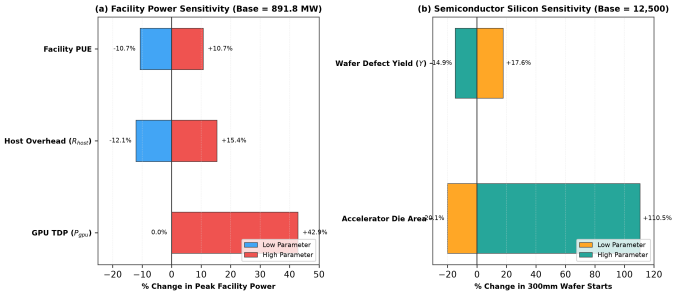



*Fig. 4. One-at-a-Time Sensitivity Tornado Analysis (500k GPUs).*

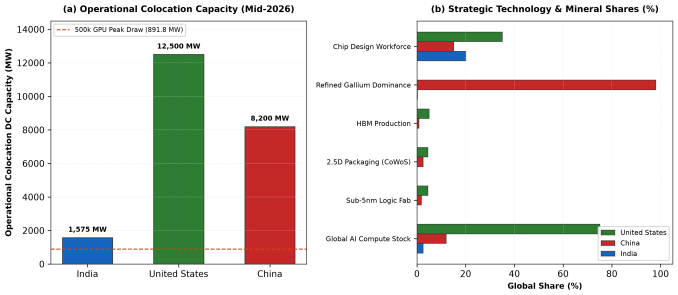



*Fig. 5. Cross-National Asymmetric Dependency Benchmarks.*

**E. Cross-National Comparative Benchmarking:** Table IV compares India, the United States, and China across strictly comparable empirical indicators. While India houses ~20% of global chip design engineers in foreign GCCs, it possesses 0% domestic sub-5nm logic fabrication, 0% CoWoS advanced packaging, and 0% HBM manufacturing.

**TABLE IV: CROSS-NATIONAL BENCHMARKING (MID-2026)**

| Indicator Name | Unit | India | US | China |
|---|---|---|---|---|
| Colocation DC Capacity | MW | 1,575 | 12,500 | 8,200 |
| Global AI Compute Share | % | <3.0% | 75.0% | 12.0% |
| Sub-5nm Logic Fab Share | % | 0.0% | <5.0% | <2.0% |
| 2.5D CoWoS Pkg Share | % | 0.0% | <5.0% | <3.0% |
| HBM Production Share | % | 0.0% | ~5.0% | <1.0% |
| Refined Gallium Dominance | % | <0.5% | 0.0% | 98.0% |
| Average Colocation PUE | ratio | 1.55 | 1.22 | 1.38 |
| Design Talent Workforce | % | ~20% | ~35% | ~15% |

## V. STRATEGIC ROADMAP & POLICY IMPLICATIONS

**A. Documented Priority Inversion: Debunking 'Foundry-First':** Standard policy discourse advocates spending $15–$20 billion to build a domestic sub-5nm logic foundry. Our cascade model proves this is a catastrophic strategic misallocation. A 500k-GPU sovereign cluster consumes only 12,500 wafers/year—less than 3% of the 480,000 annual wafer volume needed to economically sustain a modern gigafab. Furthermore, domestic 3nm logic dies produce zero operational accelerators without 2.5D packaging, HBM, and 891.8 MW of substation headroom.

The framework directs an **immediate priority inversion**:

- *Old Priority:* Sub-5nm logic fabrication megaprojects ($15B+ capital risk, 7–10 year lead time).
- *Inverted New Priority:* High-voltage Green AI transmission corridors, spatial decentralization to power-surplus states, and bilateral packaging/memory joint ventures.

**B. Phased 10-Year Implementation Plan:** Interventions are prioritized into three operational categories based on multi-criteria feasibility and sovereignty leverage:

**1) Category A: Build Now (2026–2028) — Immediate Control:**

• *Spatial Diversification:* Institute 15–20% capital rebates directing greenfield AI clusters away from Mumbai into Gujarat, Rajasthan, Odisha, and Madhya Pradesh, reducing Mumbai's concentration below 35%.

• *Direct Liquid Cooling Mandate:* Condition IndiaAI compute vouchers on PUE ≤ 1.25 and closed-loop dry/adiabatic heat rejection, eliminating 11.25 billion litres/year of freshwater loss.

• *Green Grid Corridors:* Amend Central Electricity Authority (CEA) regulations to create a fast-track 9-month approval window for dedicated 400/220 kV substations serving compute clusters ≥ 50 MW.

**2) Category B: Build through Partnership (2028–2031):**

• *Advanced Packaging JVs:* Expand approved domestic ATMP facilities (Tata Electronics, Micron) from mature wire-bond toward 2.5D silicon and glass interposers via joint ventures with Taiwanese and Japanese OSATs under Semicon 2.0.

• *Gallium Refining Commercialization:* Scale CSIR-IMMT technology to extract refined gallium from domestic bauxite refinery residue (red mud) at NALCO and Hindalco, producing 50 metric tons/year of high-purity gallium.

• *Sovereign Chiplet Consortium:* Mobilize India's 120,000 GCC VLSI design engineers into an open-standard (UCIe) consortium to tape out domain-specific RISC-V matrix acceleration chiplets on mature foundry nodes.

**3) Category C: Depend and Ally (2031–2035):**

• Secure multi-year sovereign foundry capacity reservation agreements at TSMC and bilateral memory supply treaties with South Korea, leveraging India's massive automotive and consumer electronics market as strategic trade offsets.

## VI. CONCLUSION AND LIMITATIONS

This paper presents the first reproducible, multi-layer demand-cascade framework linking national AI accelerator targets directly to electric transmission, cooling thermodynamics, and semiconductor supply chains. The empirical findings prove that India's sovereign compute bottleneck is not front-end lithography volume, but 2.5D CoWoS packaging, HBM allocation, and metropolitan grid congestion. By prioritizing transmission corridors, spatial dispersion, and packaging alliances, India can deploy resilient sovereign AI infrastructure without falling victim to capital misallocations.

*Limitations:* The model assumes static annual cluster utilization (80%) without dynamic nodal power dispatch, and relies on public foundry defect density disclosures. Future research will integrate real-time AC power-flow simulations and dynamic diurnal inference scaling.

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